

Smart Transit Network System

Network Design — Final Report

A six-site routed network for a smart public transit system, addressed with VLSM from a single /16 and running a mixture of RIPv2 and five different kinds of static routing. This report contains the VLSM tree, the complete IP address table, the configuration commands for all six routers, and the assumptions the design rests on.

BASE NETWORK ADDRESS

18.18.0.0 / 16

Four-member group — sum of the last three digits of the fourth member's ID: 7 + 8 + 3 = 18

Student ID	Name	Assigned project tasks
22201828	Mahibi Islam	Base IP / VLSM calculations; CCH & DPC router interface and DHCP configuration
22101150	MD. Mubin Ibne Azad	PSC web, email and DHCP service setup; CCH DNS server configuration
22301281	Pritom Chakraborty	Advanced routing implementation (RIPv2, floating route, recursive route)
22201783	Abdullah Al Muqit	Subnet & topology diagram labelling; failure-scenario testing and report compilation

Contents

- 1 Base Network Address
- 2 Host Requirements and Subnet Sizing
- 3 Subnet Allocation Table
- 4 VLSM Tree Diagram
- 5 IP Address Table
- 6 Router Configuration Commands
- 7 Server, DHCP and End-Device Configuration
- 8 Routing Design
- 9 Assumptions
- 10 Verification

Deliverables covered by this report

#	Deliverable	Where it is
1	Work distribution among group members	Cover page
2	Packet Tracer file (.pkt)	2310_Smart_Transit_Network.pkt, submitted alongside this report
3	Topology diagram with subnet labels	Labeled_Topology.png, with the subnet labels listed in section 3 and the per-link addressing in Table 5
4	VLSM tree diagram	Section 4, Figures 2a and 2b
5	IP address table (all devices and interfaces)	Section 5, and section 7 for the server settings
6	Configuration commands for all routers	Section 6

1. Base Network Address

The project derives the base network from the student IDs of the group. This is a four-member group, so the governing rule is: *"4-member group — use the sum of the last 3 digits of the fourth member's ID"*. The derived value is used for both of the first two octets, the remaining octets are zero, and the prefix is /16.

Step	Working	Value
Group size	Four members, so rule 3 applies	4 members
Fourth member	Abdullah Al Muqit	22201783
Last three digits	2 2 2 0 1 7 8 3	7, 8, 3
Sum of those digits	7 + 8 + 3	18
First two octets	The derived value is used twice	18.18

Step	Working	Value
Base network	Remaining octets are 0.0 with a /16 prefix	18.18.0.0/16

Result — every address in this document is taken from 18.18.0.0/16. The block holds 65,536 addresses (65,534 usable), which is far more than the 740 devices the six centres require, so the whole design fits comfortably inside one /16 and still leaves room for growth.

2. Host Requirements and Subnet Sizing

Each centre holds one router, one internal switch, two PCs and one printer, plus the servers listed in the specification. The *Devices* column is the figure given in the project brief and is treated as the number of hosts the LAN must be able to address. One extra address is added for the router interface that acts as the default gateway, and the prefix chosen is the smallest one that still covers that total.

Table 1 — Sizing calculation for the six LAN subnets.

Centre	Full name	Devices	Hosts required	Host bits	Usable addresses	Prefix	Subnet mask	Spare
CCH	Central Control Hub	200	201	8	254	/24	255.255.255.0	54
DPC	Depot Center	150	151	8	254	/24	255.255.255.0	104
PSC	Passenger Service Center	135	136	8	254	/24	255.255.255.0	119
FDU	Fleet Dispatch Unit	120	121	7	126	/25	255.255.255.128	6
DAU	Data Analytics Unit	85	86	7	126	/25	255.255.255.128	41
RSP	Regional Stop Point	50	51	6	62	/26	255.255.255.192	12

Because the six LANs need six different sizes (254, 254, 254, 126, 126 and 62 usable addresses), a single fixed-length mask would waste a large part of the block. Variable Length Subnet Masking is therefore used: the largest requirement is allocated first and each smaller requirement is carved out of the space that remains.

2.1 Point-to-point and shared links

Six further subnets are needed for the links between the routers.

Table 2 — Sizing calculation for the inter-router links.

Link	Type	Hosts required	Host bits	Usable addresses	Prefix	Subnet mask
CCH ↔ PSC	Shared switched segment (common switch)	2	3	6	/29	255.255.255.248
CCH ↔ DPC	Serial point-to-point link	2	2	2	/30	255.255.255.252
DPC ↔ FDU	Serial point-to-point link	2	2	2	/30	255.255.255.252
FDU ↔ RSP	Serial point-to-point link	2	2	2	/30	255.255.255.252
PSC ↔ DAU	Serial point-to-point link	2	2	2	/30	255.255.255.252
DPC ↔ PSC	Backup serial link (floating static only)	2	2	2	/30	255.255.255.252

Why the CCH–PSC segment gets a /29 and not a /30. CCH and PSC are joined through a common shared switch rather than a direct cable, so that segment is a single broadcast domain that could hold more than two devices. It is given a /29 (six usable addresses) so that a switch management address or a third router can be added later without renumbering. Every other link is a direct point-to-point serial cable that can never hold more than two addresses, so each gets a /30.

3. Subnet Allocation Table

The twelve subnets below are allocated largest-first and are completely contiguous: every block starts at the next correctly aligned boundary after the previous one, so no address space is stranded between them.

Table 3 — The complete VLSM allocation derived from 18.18.0.0/16.

Subnet	Network address	Prefix	Subnet mask	Usable host range	Broadcast	Usable	Needed
LAN subnets							
CCH LAN	18.18.0.0	/24	255.255.255.0	18.18.0.1 – 18.18.0.254	18.18.0.255	254	200
DPC LAN	18.18.1.0	/24	255.255.255.0	18.18.1.1 – 18.18.1.254	18.18.1.255	254	150
PSC LAN	18.18.2.0	/24	255.255.255.0	18.18.2.1 – 18.18.2.254	18.18.2.255	254	135
FDU LAN	18.18.3.0	/25	255.255.255.128	18.18.3.1 – 18.18.3.126	18.18.3.127	126	120
DAU LAN	18.18.3.128	/25	255.255.255.128	18.18.3.129 – 18.18.3.254	18.18.3.255	126	85
RSP LAN	18.18.4.0	/26	255.255.255.192	18.18.4.1 – 18.18.4.62	18.18.4.63	62	50
Inter-router subnets							
CCH ↔ PSC	18.18.4.64	/29	255.255.255.248	18.18.4.65 – 18.18.4.70	18.18.4.71	6	2
CCH ↔ DPC	18.18.4.72	/30	255.255.255.252	18.18.4.73 – 18.18.4.74	18.18.4.75	2	2
DPC ↔ FDU	18.18.4.76	/30	255.255.255.252	18.18.4.77 – 18.18.4.78	18.18.4.79	2	2
FDU ↔ RSP	18.18.4.80	/30	255.255.255.252	18.18.4.81 – 18.18.4.82	18.18.4.83	2	2
PSC ↔ DAU	18.18.4.84	/30	255.255.255.252	18.18.4.85 – 18.18.4.86	18.18.4.87	2	2
DPC ↔ PSC	18.18.4.88	/30	255.255.255.252	18.18.4.89 – 18.18.4.90	18.18.4.91	2	2

Table 4 — Address space utilisation.

Measure	Value
Total addresses in the base block 18.18.0.0/16	65,536
Addresses consumed by the twelve subnets	1,116
Highest address used in the design	18.18.4.91
Utilisation of the base block	1.70 %
Contiguous space still free (18.18.4.92 onwards)	64,420

4. VLSM Tree Diagram

Each split divides a block into two equal halves, so a /16 becomes two /17s, a /24 becomes two /25s, and so on. A branch stops as soon as the block is small enough for the requirement it serves. Figure 2a shows the first level of the design; the 18.18.4.0/24 branch is expanded separately in Figure 2b because it goes six levels deeper.

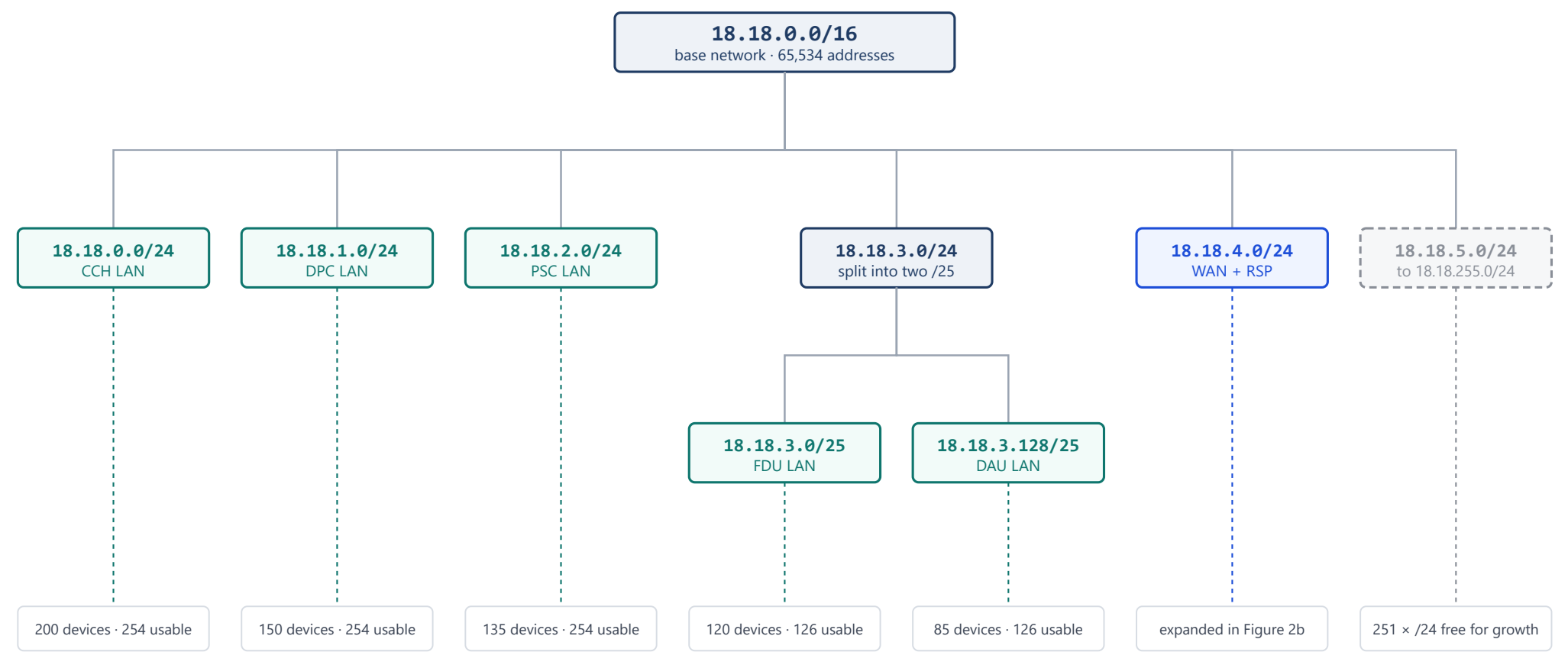


Figure 2a — Top level of the VLSM tree: 18.18.0.0/16 divided into the /24 blocks, with 18.18.3.0/24 split once more to produce the FDU and DAU LANs.

■ LAN subnet ■ Inter-router subnet ■ Block that is split further ■ Unallocated (free)

4.1 Expansion of the 18.18.4.0/24 branch

This branch carries the smallest LAN and all six inter-router subnets, so it is halved repeatedly down to /30. Reading left to right, every parent block is divided into the two halves shown to its right; a block that is used stops there and its purpose is given on the right-hand side.

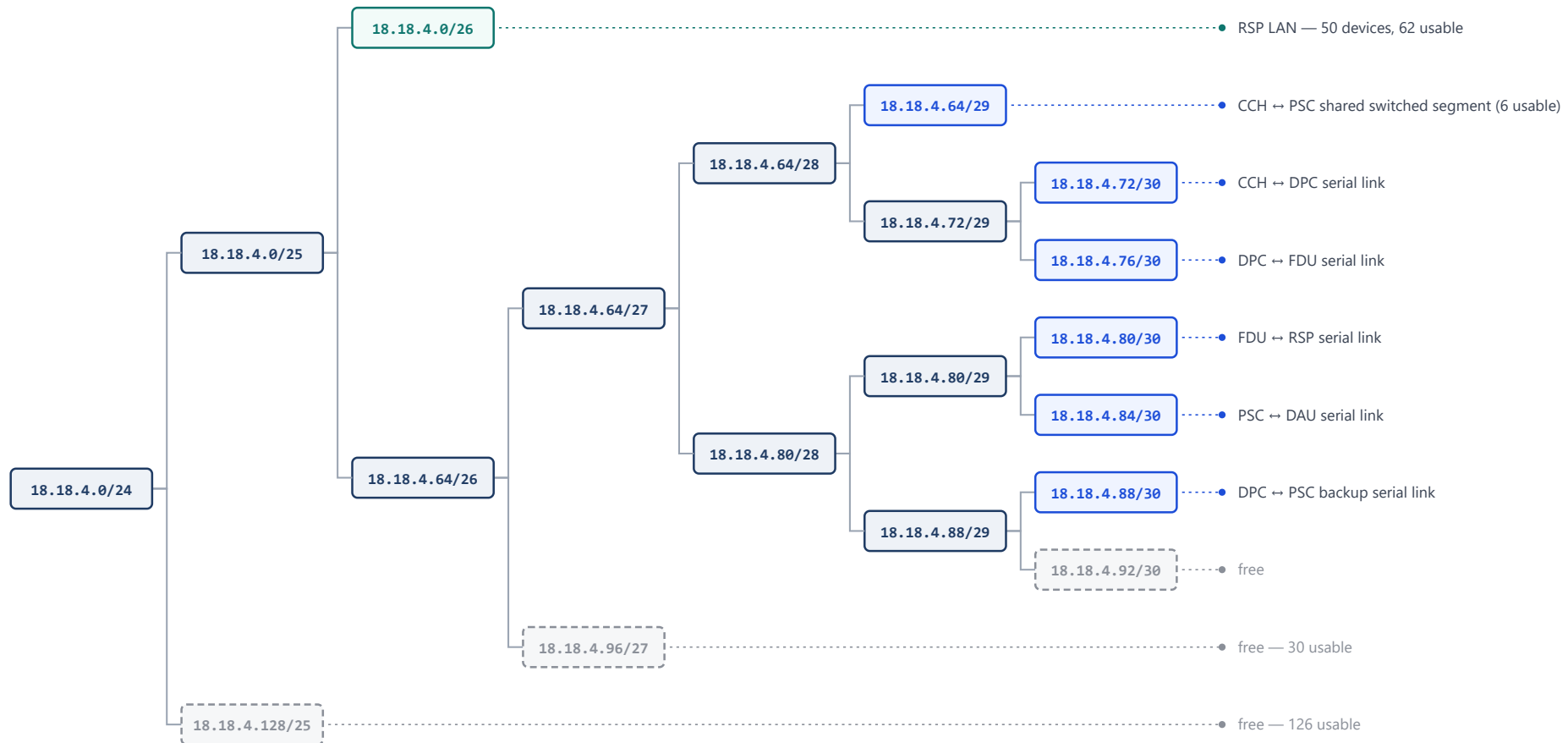


Figure 2b — Binary expansion of 18.18.4.0/24 from /24 down to /30. Free blocks are shown with a dashed outline.

■ LAN subnet ■ Inter-router subnet ■ Block that is split further ■ Unallocated (free)

5. IP Address Table

Every interface of every device in the topology is listed below, grouped by centre. The addresses marked *DHCP* are the first addresses each pool hands out; because a DHCP server allocates in the order requests arrive, an individual PC may end up with a different address from the same pool.

The six access switches and the shared switch operate purely at layer 2 and forward traffic on the basis of MAC addresses, so they need no IP address for the network to work. If remote management is wanted later, an address from the reserved range .2–.9 of each LAN can be given to interface VLAN 1 together with a default gateway.

Table 5 — Inter-router link addressing at a glance.

Link	Subnet	Router A	Address A	Router B	Address B
CCH ↔ PSC (shared switch)	18.18.4.64/29	CCH	18.18.4.65	PSC	18.18.4.66
CCH ↔ DPC (serial)	18.18.4.72/30	CCH	18.18.4.73	DPC	18.18.4.74
DPC ↔ FDU (serial)	18.18.4.76/30	DPC	18.18.4.77	FDU	18.18.4.78
FDU ↔ RSP (serial)	18.18.4.80/30	FDU	18.18.4.81	RSP	18.18.4.82
PSC ↔ DAU (serial)	18.18.4.84/30	PSC	18.18.4.85	DAU	18.18.4.86
DPC ↔ PSC (backup serial)	18.18.4.88/30	DPC	18.18.4.89	PSC	18.18.4.90

5.1 CCH — Central Control Hub

LAN subnet 18.18.0.0/24 · **Default gateway** 18.18.0.1 · Static addressing only (no DHCP)

Device	Model	Interface	IPv4 address	Subnet mask	Default gateway	DNS server	Method
CCH	Router 2911	GigabitEthernet0/0	18.18.0.1	255.255.255.0	—	—	Static
CCH	Router 2911	GigabitEthernet0/1	18.18.4.65	255.255.255.248	—	—	Static
CCH	Router 2911	Serial0/3/0	18.18.4.73	255.255.255.252	—	—	Static
Switch0	Switch 2960	Layer 2 only	—	—	—	—	—
DNS Server	Server-PT	FastEthernet0	18.18.0.2	255.255.255.0	18.18.0.1	18.18.0.2	Static
Email Server	Server-PT	FastEthernet0	18.18.0.3	255.255.255.0	18.18.0.1	18.18.0.2	Static
PC0	PC-PT	FastEthernet0	18.18.0.10	255.255.255.0	18.18.0.1	18.18.0.2	Static
PC1	PC-PT	FastEthernet0	18.18.0.11	255.255.255.0	18.18.0.1	18.18.0.2	Static
Printer0	Printer-PT	FastEthernet0	18.18.0.12	255.255.255.0	18.18.0.1	—	Static

5.2 DPC — Depot Center

LAN subnet 18.18.1.0/24 · **Default gateway** 18.18.1.1 · Router-based DHCP — pool DPC_POOL, excluded 18.18.1.1–18.18.1.10

Device	Model	Interface	IPv4 address	Subnet mask	Default gateway	DNS server	Method
DPC	Router 2911	GigabitEthernet0/0	18.18.1.1	255.255.255.0	—	—	Static
DPC	Router 2911	Serial0/3/0	18.18.4.74	255.255.255.252	—	—	Static

Device	Model	Interface	IPv4 address	Subnet mask	Default gateway	DNS server	Method
DPC	Router 2911	Serial0/3/1	18.18.4.77	255.255.255.252	—	—	Static
DPC	Router 2911	Serial0/2/0 backup	18.18.4.89	255.255.255.252	—	—	Static
Switch1	Switch 2960	Layer 2 only	—	—	—	—	—
PC2	PC-PT	FastEthernet0	18.18.1.11	255.255.255.0	18.18.1.1	18.18.0.2	DHCP
PC3	PC-PT	FastEthernet0	18.18.1.12	255.255.255.0	18.18.1.1	18.18.0.2	DHCP
Printer1	Printer-PT	FastEthernet0	18.18.1.13	255.255.255.0	18.18.1.1	18.18.0.2	DHCP

5.3 FDU — Fleet Dispatch Unit

LAN subnet 18.18.3.0/25 · **Default gateway** 18.18.3.1 · Router-based DHCP — pool FDU_POOL, excluded 18.18.3.1–18.18.3.10

Device	Model	Interface	IPv4 address	Subnet mask	Default gateway	DNS server	Method
FDU	Router 2911	GigabitEthernet0/0	18.18.3.1	255.255.255.128	—	—	Static
FDU	Router 2911	Serial0/3/1	18.18.4.78	255.255.255.252	—	—	Static
FDU	Router 2911	Serial0/3/0	18.18.4.81	255.255.255.252	—	—	Static
Switch2	Switch 2960	Layer 2 only	—	—	—	—	—
PC4	PC-PT	FastEthernet0	18.18.3.11	255.255.255.128	18.18.3.1	18.18.0.2	DHCP
PC5	PC-PT	FastEthernet0	18.18.3.12	255.255.255.128	18.18.3.1	18.18.0.2	DHCP
Printer2	Printer-PT	FastEthernet0	18.18.3.13	255.255.255.128	18.18.3.1	18.18.0.2	DHCP

5.4 PSC — Passenger Service Center

LAN subnet 18.18.2.0/24 · **Default gateway** 18.18.2.1 · Dedicated DHCP server at 18.18.2.4 (not on the router), pool starts at 18.18.2.11 · the web server hosts www.transitline.local

Device	Model	Interface	IPv4 address	Subnet mask	Default gateway	DNS server	Method
PSC	Router 2911	GigabitEthernet0/0	18.18.2.1	255.255.255.0	—	—	Static
PSC	Router 2911	GigabitEthernet0/1	18.18.4.66	255.255.255.248	—	—	Static
PSC	Router 2911	Serial0/3/0	18.18.4.85	255.255.255.252	—	—	Static
PSC	Router 2911	Serial0/3/1 backup	18.18.4.90	255.255.255.252	—	—	Static
Switch3	Switch 2960	Layer 2 only	—	—	—	—	—
Web Server	Server-PT	FastEthernet0	18.18.2.2	255.255.255.0	18.18.2.1	18.18.0.2	Static
Email Server	Server-PT	FastEthernet0	18.18.2.3	255.255.255.0	18.18.2.1	18.18.0.2	Static
DHCP Server	Server-PT	FastEthernet0	18.18.2.4	255.255.255.0	18.18.2.1	18.18.0.2	Static
PC6	PC-PT	FastEthernet0	18.18.2.11	255.255.255.0	18.18.2.1	18.18.0.2	DHCP
PC7	PC-PT	FastEthernet0	18.18.2.12	255.255.255.0	18.18.2.1	18.18.0.2	DHCP
Printer3	Printer-PT	FastEthernet0	18.18.2.13	255.255.255.0	18.18.2.1	18.18.0.2	DHCP

5.5 DAU — Data Analytics Unit

LAN subnet 18.18.3.128/25 · **Default gateway** 18.18.3.129 · Static addressing only (no DHCP)

Device	Model	Interface	IPv4 address	Subnet mask	Default gateway	DNS server	Method
DAU	Router 2911	GigabitEthernet0/0	18.18.3.129	255.255.255.128	—	—	Static
DAU	Router 2911	Serial0/3/0	18.18.4.86	255.255.255.252	—	—	Static
Switch4	Switch 2960	Layer 2 only	—	—	—	—	—
PC8	PC-PT	FastEthernet0	18.18.3.130	255.255.255.128	18.18.3.129	18.18.0.2	Static
PC9	PC-PT	FastEthernet0	18.18.3.131	255.255.255.128	18.18.3.129	18.18.0.2	Static
Printer4	Printer-PT	FastEthernet0	18.18.3.132	255.255.255.128	18.18.3.129	—	Static

5.6 RSP — Regional Stop Point

LAN subnet 18.18.4.0/26 · **Default gateway** 18.18.4.1 · Static addressing only (no DHCP)

Device	Model	Interface	IPv4 address	Subnet mask	Default gateway	DNS server	Method
RSP	Router 2911	GigabitEthernet0/0	18.18.4.1	255.255.255.192	—	—	Static
RSP	Router 2911	Serial0/3/0	18.18.4.82	255.255.255.252	—	—	Static
Switch5	Switch 2960	Layer 2 only	—	—	—	—	—
PC10	PC-PT	FastEthernet0	18.18.4.10	255.255.255.192	18.18.4.1	18.18.0.2	Static
PC11	PC-PT	FastEthernet0	18.18.4.11	255.255.255.192	18.18.4.1	18.18.0.2	Static
Printer5	Printer-PT	FastEthernet0	18.18.4.12	255.255.255.192	18.18.4.1	—	Static

6. Router Configuration Commands

The listings below are the complete configurations, in the order they should be pasted into each router's CLI. Answer no to the initial setup dialogue first. Lines marked **REQUIRED** in red are the five static route types the project asks for; green lines are explanatory comments and can be pasted safely.

Order of work. Configure all interfaces on all six routers first and confirm that every serial link comes up (show ip interface brief), then apply the routing sections. RIPv2 needs about 30–45 seconds to converge before the routing tables settle.

6.1 CCH — Central Control Hub

Mixed routing: static routes to every LAN plus RIPv2 with static redistribution

```
enable
configure terminal
hostname CCH
no ip domain-lookup
!
!--- LAN interface (static addressing only, no DHCP) ---
interface GigabitEthernet0/0
description CCH LAN 18.18.0.0/24
ip address 18.18.0.1 255.255.255.0
no shutdown
exit
!
!--- Shared switched segment with PSC ---
interface GigabitEthernet0/1
description Shared segment CCH-PSC 18.18.4.64/29
ip address 18.18.4.65 255.255.255.248
no shutdown
exit
!
!--- Serial link to DPC ---
interface Serial0/3/0
description WAN CCH-DPC 18.18.4.72/30
ip address 18.18.4.73 255.255.255.252
clock rate 64000
no shutdown
exit
!
!=== STATIC ROUTES =====
! [REQUIRED] Next-hop static route to the FDU LAN, next hop = DPC
ip route 18.18.3.0 255.255.255.128 18.18.4.74
! Static route to the DAU LAN, next hop = PSC
ip route 18.18.3.128 255.255.255.128 18.18.4.66
! Static route to the RSP LAN, next hop = DPC
ip route 18.18.4.0 255.255.255.192 18.18.4.74
! Static routes to the remaining LANs so that CCH reaches
! every network in the design manually as well as by RIPv2
ip route 18.18.1.0 255.255.255.0 18.18.4.74
ip route 18.18.2.0 255.255.255.0 18.18.4.66
!
!=== RIPv2 + REDISTRIBUTION =====
router rip
version 2
no auto-summary
network 18.0.0.0
passive-interface GigabitEthernet0/0
redistribute static
default-metric 1
exit
!
end
write memory
```

6.2 DPC — Depot Center

Router-based DHCP, RIPv2, and the floating static route to PSC over the backup link

```
enable
configure terminal
hostname DPC
no ip domain-lookup
!
!--- LAN interface ---
interface GigabitEthernet0/0
  description DPC LAN 18.18.1.0/24
  ip address 18.18.1.1 255.255.255.0
  no shutdown
exit
!
!--- Serial link to CCH (primary path) ---
interface Serial0/3/0
  description WAN DPC-CCH 18.18.4.72/30
  ip address 18.18.4.74 255.255.255.252
  no shutdown
exit
!
!--- Serial link to FDU ---
interface Serial0/3/1
  description WAN DPC-FDU 18.18.4.76/30
  ip address 18.18.4.77 255.255.255.252
  clock rate 64000
  no shutdown
exit
!
!--- Backup serial link to PSC (carries the floating static route only) ---
interface Serial0/2/0
  description BACKUP DPC-PSC 18.18.4.88/30
  ip address 18.18.4.89 255.255.255.252
  clock rate 64000
  no shutdown
exit
!
!=== ROUTER-BASED DHCP FOR THE DPC LAN =====
ip dhcp excluded-address 18.18.1.1 18.18.1.10
ip dhcp pool DPC_POOL
  network 18.18.1.0 255.255.255.0
  default-router 18.18.1.1
  dns-server 18.18.0.2
exit
!
!=== FLOATING STATIC ROUTE =====
! [REQUIRED] Administrative distance 130 is higher than the 120 used by
! RIPv2, so this route stays dormant in the routing table and installs
! only when the dynamic path to the PSC LAN disappears.
ip route 18.18.2.0 255.255.255.0 18.18.4.90 130
!
!=== RIPv2 =====
router rip
  version 2
  no auto-summary
  network 18.0.0.0
  passive-interface GigabitEthernet0/0
  ! The backup link must not carry RIPv2. If it did, DPC would learn the
  ! PSC LAN in a single hop across it, the floating route would never
  ! install, and the failure scenario could not be demonstrated.
  passive-interface Serial0/2/0
exit
!
end
write memory
```

6.3 FDU — Fleet Dispatch Unit

Router-based DHCP, RIPv2, and the static route into the non-RIP RSP site

```
enable
configure terminal
hostname FDU
no ip domain-lookup
!
!--- LAN interface ---
interface GigabitEthernet0/0
  description FDU LAN 18.18.3.0/25
  ip address 18.18.3.1 255.255.255.128
  no shutdown
exit
!
!--- Serial link to DPC ---
interface Serial0/3/1
  description WAN FDU-DPC 18.18.4.76/30
  ip address 18.18.4.78 255.255.255.252
  no shutdown
exit
!
!--- Serial link to RSP ---
interface Serial0/3/0
  description WAN FDU-RSP 18.18.4.80/30
  ip address 18.18.4.81 255.255.255.252
  clock rate 64000
  no shutdown
exit
!
!=== ROUTER-BASED DHCP FOR THE FDU LAN ===
ip dhcp excluded-address 18.18.3.1 18.18.3.10
ip dhcp pool FDU_POOL
  network 18.18.3.0 255.255.255.128
  default-router 18.18.3.1
  dns-server 18.18.0.2
exit
!
!=== STATIC ROUTE ===
! RSP does not run RIPv2, so its LAN is reached through a next-hop
! static route which is then injected into RIPv2 by the redistribution
! statement below, making the RSP LAN visible to DPC, CCH and PSC.
ip route 18.18.4.0 255.255.255.192 18.18.4.82
!
!=== RIPv2 + REDISTRIBUTION ===
router rip
  version 2
  no auto-summary
  network 18.0.0.0
  passive-interface GigabitEthernet0/0
  ! RSP must not receive RIPv2 updates
  passive-interface Serial0/3/0
  redistribute static
  default-metric 1
exit
!
end
write memory
```

6.4 PSC — Passenger Service Center

RIPv2, the recursive static route to RSP, and the floating static route back to DPC

```
enable
configure terminal
hostname PSC
no ip domain-lookup
!
!--- LAN interface (hosts are served by the dedicated DHCP server) ---
interface GigabitEthernet0/0
  description PSC LAN 18.18.2.0/24
  ip address 18.18.2.1 255.255.255.0
  no shutdown
exit
!
!--- Shared switched segment with CCH ---
interface GigabitEthernet0/1
  description Shared segment PSC-CCH 18.18.4.64/29
  ip address 18.18.4.66 255.255.255.248
  no shutdown
exit
!
!--- Serial link to DAU ---
interface Serial0/3/0
  description WAN PSC-DAU 18.18.4.84/30
  ip address 18.18.4.85 255.255.255.252
  clock rate 64000
  no shutdown
exit
!
!--- Backup serial link to DPC (carries the floating static route only) ---
interface Serial0/3/1
  description BACKUP PSC-DPC 18.18.4.88/30
  ip address 18.18.4.90 255.255.255.252
  no shutdown
exit
!
!=== STATIC ROUTES =====
! DAU does not run RIPv2, so its LAN is reached by a next-hop static route
ip route 18.18.3.128 255.255.255.128 18.18.4.86
! [REQUIRED] Recursive static route to the RSP LAN. The next hop is CCH,
! which is not directly attached to RSP, so the router must resolve the
! next hop against its own table: PSC -> CCH -> DPC -> FDU -> RSP.
ip route 18.18.4.0 255.255.255.192 18.18.4.65
! [REQUIRED] Floating static route back to the DPC LAN over the backup link
ip route 18.18.1.0 255.255.255.0 18.18.4.89 130
!
!=== RIPv2 + REDISTRIBUTION =====
router rip
  version 2
  no auto-summary
  network 18.0.0.0
  passive-interface GigabitEthernet0/0
  ! DAU must not receive RIPv2 updates
  passive-interface Serial0/3/0
  ! The backup link must not carry RIPv2 (see the note on DPC)
  passive-interface Serial0/3/1
  redistribute static
  default-metric 1
exit
!
end
write memory
```

6.5 DAU — Data Analytics Unit

No RIPv2. Static addressing and one static route for every remote network

```
enable
configure terminal
hostname DAU
no ip domain-lookup
!
!--- LAN interface (static addressing only, no DHCP) ---
interface GigabitEthernet0/0
description DAU LAN 18.18.3.128/25
ip address 18.18.3.129 255.255.255.128
no shutdown
exit
!
!--- Serial link to PSC ---
interface Serial0/3/0
description WAN DAU-PSC 18.18.4.84/30
ip address 18.18.4.86 255.255.255.252
no shutdown
exit
!
!=== STATIC ROUTES =====
! Summary static routes are not allowed and only RSP may hold a default
! route, so every remote LAN is listed individually with PSC as next hop.
ip route 18.18.0.0 255.255.255.0 18.18.4.85
ip route 18.18.1.0 255.255.255.0 18.18.4.85
ip route 18.18.2.0 255.255.255.0 18.18.4.85
ip route 18.18.3.0 255.255.255.128 18.18.4.85
! [REQUIRED] Recursive static route to the RSP LAN with PSC as next hop.
! PSC in turn holds its own recursive route through CCH, DPC and FDU.
ip route 18.18.4.0 255.255.255.192 18.18.4.85
!
!--- Transit links, so that DAU can also reach every router interface ---
ip route 18.18.4.64 255.255.255.248 18.18.4.85
ip route 18.18.4.72 255.255.255.252 18.18.4.85
ip route 18.18.4.76 255.255.255.252 18.18.4.85
ip route 18.18.4.80 255.255.255.252 18.18.4.85
ip route 18.18.4.88 255.255.255.252 18.18.4.85
!
end
write memory
```

6.6 RSP — Regional Stop Point

No RIPv2. The exit-interface static route and the only default route in the network

```
enable
configure terminal
hostname RSP
no ip domain-lookup
!
!--- LAN interface (static addressing only, no DHCP) ---
interface GigabitEthernet0/0
description RSP LAN 18.18.4.0/26
ip address 18.18.4.1 255.255.255.192
no shutdown
exit
!
!--- Serial link to FDU ---
interface Serial0/3/0
description WAN RSP-FDU 18.18.4.80/30
ip address 18.18.4.82 255.255.255.252
no shutdown
exit
!
!=== STATIC ROUTES =====
! [REQUIRED] Exit-interface static route to the CCH LAN. Only the outgoing
! interface is given and no next-hop address, so the route is not
! "fully specified" and stays within the rules of the project.
ip route 18.18.0.0 255.255.255.0 Serial0/3/0
! [REQUIRED] Default static route towards FDU. RSP is the only router in
! the whole design that is permitted to hold a 0.0.0.0/0 route.
ip route 0.0.0.0 0.0.0.0 18.18.4.81
!
end
write memory
```

7. Server, DHCP and End-Device Configuration

The services below are configured through the Packet Tracer GUI rather than the router CLI, so they are given as settings tables.

7.1 DNS server — CCH, 18.18.0.2

Open the server, go to **Services** > **DNS**, set the service to **On** and add the following A records. Only the first record is required by the specification; the other two are added so that the mail servers can also be reached by name.

Name	Type	Address	Purpose
www.transitline.local	A	18.18.2.2	Web server at PSC — REQUIRED
mail.transitline.local	A	18.18.2.3	PSC email server / relay
cchmail.transitline.local	A	18.18.0.3	CCH email server

7.2 Web server — PSC, 18.18.2.2

Go to **Services** > **HTTP** and leave both HTTP and HTTPS **On**. The default `index.html` can be edited to identify the site. Once the DNS record above exists, any PC in the network can open `http://www.transitline.local` from its web browser.

7.3 Dedicated DHCP server — PSC, 18.18.2.4

Go to **Services** > **DHCP**, set the service to **On**, edit the `serverPool` entry with the values below and press **Save**.

Setting	Value
Pool name	serverPool
Default gateway	18.18.2.1
DNS server	18.18.0.2
Start IP address	18.18.2.11
Subnet mask	255.255.255.0
Maximum number of users	200

The DHCP server sits inside the PSC LAN, in the same broadcast domain as the PSC clients, so the DHCP discover messages reach it directly. No `ip helper-address` is needed on the PSC router, and the PSC router deliberately has **no** `ip dhcp pool` of its own — two DHCP sources on one LAN would hand out conflicting addresses.

7.4 Email servers — CCH 18.18.0.3 and PSC 18.18.2.3

On each server open **Services** > **EMAIL**, switch SMTP and POP3 **On** and set the domain name to `transitline.local`, then create the user accounts. PSC acts as the mail relay for the whole network, so every centre other than CCH uses the PSC server for both incoming and outgoing mail.

Centre	Incoming mail server (POP3)	Outgoing mail server (SMTP)	Example account
CCH	18.18.0.3	18.18.2.3 relay	cch@transitline.local
DPC	18.18.2.3	18.18.2.3	dpc@transitline.local
FDU	18.18.2.3	18.18.2.3	fdu@transitline.local

Centre	Incoming mail server (POP3)	Outgoing mail server (SMTP)	Example account
PSC	18.18.2.3	18.18.2.3	psc@transitline.local
DAU	18.18.2.3	18.18.2.3	dau@transitline.local
RSP	18.18.2.3	18.18.2.3	rsp@transitline.local

7.5 PCs and printers

Static hosts are set from [Desktop > IP Configuration](#) on a PC, or from [Config > INTERFACE > FastEthernet0](#) on a printer, where the default gateway is set separately under [Config > GLOBAL > Settings](#). DHCP hosts only need the **DHCP** radio button; address, mask, gateway and DNS server all arrive from the pool. The exact values for every device are in section 5.

8. Routing Design

8.1 The five required static route types

Table 6 — Where each required route type is implemented and why it qualifies as that type.

Route type	Router	Command	Why it is that type
1 Next-hop static	CCH	<code>ip route 18.18.3.0 255.255.255.128 18.18.4.74</code>	Reaches the FDU LAN by naming only the next-hop address 18.18.4.74, which is DPC's serial interface. No exit interface is given, so the route is a pure next-hop route and is not fully specified.
2 Exit-interface static	RSP	<code>ip route 18.18.0.0 255.255.255.0 Serial0/3/0</code>	Reaches the CCH LAN by naming only the outgoing interface that faces FDU. Because the link is point-to-point serial there is no ARP lookup to resolve, so an exit-interface route works without a next-hop address.
3 Recursive static	DAU	<code>ip route 18.18.4.0 255.255.255.192 18.18.4.85</code>	Reaches the RSP LAN with PSC as the next hop. PSC is not attached to RSP, so the router has to resolve the next hop against its own table before it can forward — that second lookup is what makes the route recursive.
3 Recursive static (second half)	PSC	<code>ip route 18.18.4.0 255.255.255.192 18.18.4.65</code>	Completes the chain the specification asks for. PSC hands the traffic to CCH, CCH has its own static route to the RSP LAN through DPC, and DPC reaches FDU, giving PSC → CCH → DPC → FDU → RSP.
4 Floating static	DPC	<code>ip route 18.18.2.0 255.255.255.0 18.18.4.90 130</code>	Reaches the PSC LAN across the backup link with an administrative distance of 130. RIPv2 has a distance of 120, so while the dynamic path is alive the floating route stays out of the routing table and only installs when that path is lost.
4 Floating static (return path)	PSC	<code>ip route 18.18.1.0 255.255.255.0 18.18.4.89 130</code>	The mirror image on PSC, so that traffic can flow in both directions over the backup link during a failure instead of only outbound.
5 Default static	RSP	<code>ip route 0.0.0.0 0.0.0.0 18.18.4.81</code>	Sends everything RSP does not know about to FDU. RSP is a stub site with one uplink, and it is the only router in the design that carries a 0.0.0.0/0 route.

8.2 Why RIPv2 is kept off three of the links

18.18.0.0/16 sits inside the class A network 18.0.0.0, so IOS rewrites any `network 18.18.x.0` statement as `network 18.0.0.0` and switches RIPv2 on for *every* interface in the 18.x range. That is convenient, but it activates RIPv2 on three links where it must not run:

- **The DPC–PSC backup link.** This is the important one. If RIPv2 ran across it, DPC would learn the PSC LAN in a single hop directly over the backup, with a distance of 120. The floating static route at distance 130 would then never install, the backup link would quietly become the primary path, and the failure scenario the project is built around could not be demonstrated at all. `passive-interface` on both ends stops the updates while still advertising the link itself to the rest of the RIPv2 domain.
- **FDU’s link to RSP and PSC’s link to DAU.** The specification says RIPv2 must not be enabled on DAU or RSP, so sending them RIPv2 updates would work against the intent even though neither router would answer.
- **The LAN interfaces.** Made passive as good practice: there is no router on any LAN to talk to, so broadcasting a routing table onto them only wastes bandwidth and exposes the topology to the hosts.

8.3 How the network stays fully reachable

DAU and RSP sit outside the RIPv2 domain, so their LANs would be invisible to the other routers. Two next-hop static routes fix that — one on PSC pointing at the DAU LAN and one on FDU pointing at the RSP LAN — and `redistribute static` on those routers injects them into RIPv2. CCH does the same for the routes it holds. In the other direction, DAU lists every remote network individually because summary routes are banned and it may not hold a default route, while RSP needs only its default route plus the one exit-interface route the project requires.

8.4 Compliance with the routing rules

Table 7 — Each routing rule in the specification and where it is satisfied.

Rule from the specification	How the design satisfies it	Status
RIPv2 runs on CCH, DPC, FDU and PSC	<code>router rip / version 2</code> on exactly those four routers	Met
RIPv2 does not run on DAU or RSP	Neither router has a <code>router rip</code> block, and every RIP interface facing them is <code>passive-interface</code>	Met
CCH uses static routing and RIPv2 together	Five static routes covering every LAN, plus a full RIPv2 process	Met
CCH redistributes its static routes into RIPv2	<code>redistribute static</code> with <code>default-metric 1</code>	Met
No summary static routing	Every static route carries the exact prefix of one planned subnet; none of them aggregates two or more subnets	Met
No fully specified static routing	Every static route gives either a next-hop address or an exit interface, never both	Met
Only RSP holds a default route	<code>0.0.0.0 0.0.0.0</code> appears once in the whole design	Met
VLSM throughout	Five different prefix lengths (/24, /25, /26, /29, /30) inside one classful network, with <code>no auto-summary</code> so RIPv2 carries the masks	Met

8.5 Failure scenarios

Table 8 — The behaviour the design produces under each failure the specification lists, and how to reproduce it in Packet Tracer.

Scenario	What the network does	How to verify it
A. The primary DPC–CCH link fails	DPC loses its RIPv2 route to the PSC LAN, the route ages out, and the floating static at distance 130 installs immediately. DPC then reaches PSC across the backup link. PSC does the same in reverse for the DPC LAN.	On DPC: <code>interface Serial10/3/0 → shutdown</code> . Then show <code>ip route 18.18.2.0</code> — the entry changes from R to S with distance 130. Ping PC2 → PC6.

Scenario	What the network does	How to verify it
B. The dynamic path from DPC to PSC fails	This is the same event seen from the routing side. As long as any part of the path DPC → CCH → PSC is broken, no RIPv2 route to 18.18.2.0/24 survives on DPC and the floating route takes over automatically, with no manual action.	Shut CCH's <code>GigabitEthernet0/1</code> instead, and watch the same substitution happen on DPC.
C. CCH needs the DPC LAN while the DPC link is down	CCH's own static route to 18.18.1.0/24 is withdrawn the moment Serial0/3/0 goes down. PSC's floating route to the DPC LAN has become active by then and PSC redistributes it into RIPv2, so CCH relearns the DPC LAN through PSC.	With DPC Serial0/3/0 still shut, run <code>show ip route 18.18.1.0</code> on CCH.
D. DAU reaches RSP with no direct link	DAU never has a direct path to RSP. Its recursive static route hands the traffic to PSC, and PSC's own recursive route carries it on through CCH, DPC and FDU.	From PC8: <code>tracert 18.18.4.10</code> . The hops should read 18.18.3.129, 18.18.4.85, 18.18.4.65, 18.18.4.74, 18.18.4.78, 18.18.4.82, 18.18.4.10.
E. No router except RSP falls back to a default route	Five of the six routers report <i>Gateway of last resort is not set</i> . Any destination they do not hold a route for is simply dropped, which is what the specification demands.	<code>show ip route</code> on each router and check the gateway-of-last-resort line.

9. Assumptions

The specification fixes the topology, the device counts and the routing rules, but it leaves a number of smaller decisions open. Every such decision taken while building this network is recorded below together with the reason for it, so that the design can be checked against the brief without having to guess at intent.

Table 9 — Design assumptions and the reasoning behind each one.

#	Assumption	What was decided	Why
1	Group size and base address	The group has four members, so the fourth member's ID (22201783) is the one that drives the base network: $7 + 8 + 3 = 18$, giving 18.18.0.0/16.	The specification gives a different rule for two-, three- and four-member groups, and only the four-member rule applies here.
2	Device counts are host counts	The <i>Devices</i> figure for each centre is taken as the number of hosts that LAN must be able to address, and one extra address is added for the router interface acting as the gateway.	The brief lists a device count per centre without breaking it down, so it is read as the size the LAN has to support. Adding the gateway keeps the subnet correct even at the boundary sizes.
3	Largest-first allocation	Subnets are allocated in descending order of size, starting at 18.18.0.0, and each new block begins at the next correctly aligned boundary.	This is what keeps a VLSM plan contiguous and free of alignment gaps. Allocating a small block first would strand address space that no larger subnet could then use.
4	The CCH–PSC segment is one subnet with a /29	CCH and PSC meet on a common shared switch, which is treated as a single broadcast domain and given 18.18.4.64/29 rather than a /30.	A shared switch can hold more than two devices. Six usable addresses leave room for a switch management address or a third router without renumbering the segment.
5	All other inter-router links are point-to-point serial /30s	The five direct links are serial connections and each is given a /30.	A /30 provides exactly the two usable addresses a point-to-point link can ever need, which is the smallest allocation possible without wasting address space.
6	Interface roles are uniform	GigabitEthernet0/0 is the LAN interface on all six routers; GigabitEthernet0/1 is used only on CCH and PSC for the shared segment; all remaining links use serial interfaces.	Keeping the same role on the same port number on every router makes the address table, the configurations and the topology diagram consistent and much easier to check.
7	Backup link interfaces	The backup link uses Serial0/2/0 on DPC and Serial0/3/1 on PSC.	These are the ports the backup cable occupies in the Packet Tracer file. An earlier working note referred to Serial0/3/2 on both routers; the interfaces used throughout this document are the ones that match the built topology.
8	Servers are always statically addressed	The DNS, web, email and DHCP servers hold fixed addresses even in centres where the PCs use DHCP.	DNS records, the mail relay setting and the DHCP pool definitions all point at these servers by address, so those addresses must not change when a server restarts.
9	The first ten host addresses of each LAN are reserved	Addresses ending .1 to .10 are kept for the router, servers and future infrastructure, and the DHCP pools are configured to start at .11.	It keeps a predictable block for equipment that must be reachable at a known address, and it is why <code>ip dhcp excluded-address</code> covers the same range on the router-based pools.
10	DHCP addresses in the table are the first available leases	PC2 is shown as 18.18.1.11, PC3 as 18.18.1.12 and so on.	A DHCP server allocates in the order requests arrive, so a given PC may receive a different address from the same pool. The pool range is what is guaranteed, not the individual address.

#	Assumption	What was decided	Why
11	The PSC DHCP server needs no relay	The dedicated DHCP server sits inside the PSC LAN, and the PSC router is left with no DHCP pool of its own.	Server and clients share one broadcast domain, so the DHCP discover messages reach the server directly and no <code>ip helper-address</code> is required. Two DHCP sources on one LAN would hand out conflicting addresses.
12	RIPv2 network statements are written classfully	Each RIPv2 process uses the single statement <code>network 18.0.0.0</code> , together with no <code>auto-summary</code> .	18.18.0.0/16 belongs to class A network 18.0.0.0, and IOS rewrites any <code>network 18.18.x.0</code> into that form anyway. no <code>auto-summary</code> is essential because the design mixes /24, /25, /26, /29 and /30 inside one classful network and the masks have to travel with the updates.
13	Passive interfaces control where RIPv2 is allowed to go	The backup link, the links towards DAU and RSP, and all LAN interfaces are declared <code>passive-interface</code> .	A single classful network statement would otherwise switch RIPv2 on everywhere. The backup link is the critical case: RIPv2 across it would give DPC a one-hop route to the PSC LAN and the floating static route would never install. See section 8.2.
14	A default metric is set where statics are redistributed	CCH, FDU and PSC carry <code>default-metric 1</code> under <code>router rip</code> .	It guarantees that redistributed static routes enter the RIPv2 domain with a usable hop count instead of being seeded as unreachable, which is what makes the DAU and RSP LANs visible to the rest of the network.
15	DAU lists every remote network separately	DAU carries one static route per destination rather than a summary route or a default route.	Summary static routing is not allowed and only RSP may hold a default route, so an explicit list is the only compliant way for DAU to reach the rest of the network. The five routes for the transit links are optional and simply let DAU ping router interfaces as well as LANs.
16	The DAU failure scenario is read as an indirect-path requirement	DAU has exactly one uplink, to PSC, so if that link fails DAU is isolated and no routing can recover it. The requirement is therefore taken to mean that DAU must reach RSP without any direct link, through the recursive chain DAU → PSC → CCH → DPC → FDU → RSP.	There is no direct DAU–RSP link anywhere in the topology, and the specification itself spells out that chain and requires PSC to hold a recursive route to RSP. Scenario D in section 8.5 verifies exactly that path.
17	Switches carry no IP addressing	The six access switches and the shared switch are left as pure layer-2 devices.	They forward on MAC addresses and need no IP address to do their job. A VLAN 1 management address from the reserved .2–.9 range can be added later if remote management is wanted.
18	Clock rate is applied on the DCE end of each serial link	<code>clock rate 64000</code> appears once per serial link in the configurations.	A serial link only comes up when the DCE end supplies clocking. Which end is DCE depends on how the cable was dropped in Packet Tracer, and the command is simply ignored on a DTE end, so it is safe to leave in place.
19	Mail domain and relay behaviour	The mail domain is <code>transitline.local</code> . Every centre except CCH uses the PSC server (18.18.2.3) for both incoming and outgoing mail; CCH collects mail from its own server but still relays outgoing mail through PSC.	The brief places email servers only at CCH and PSC and states that all other units send mail through PSC, which is what a relay does.

10. Verification

10.1 Connectivity tests

Allow RIPv2 about 45 seconds to converge after the configurations are applied, then run the tests below from the Desktop > Command Prompt of each PC. The first ping of a sequence may time out while ARP resolves; the ones after it should not.

Table 10 — End-to-end connectivity tests and what each one proves.

Test	Addresses	What it proves	Expected
PC0 → PC2	18.18.0.10 → 18.18.1.11	CCH LAN to DPC LAN across the primary serial link	Reply
PC0 → PC4	18.18.0.10 → 18.18.3.11	Exercises the required next-hop static route on CCH	Reply
PC0 → PC10	18.18.0.10 → 18.18.4.10	CCH to the RSP stub site, three routers away	Reply
PC10 → PC0	18.18.4.10 → 18.18.0.10	Exercises the required exit-interface static route on RSP	Reply
PC8 → PC10	18.18.3.130 → 18.18.4.10	Exercises both recursive static routes, on DAU and on PSC	Reply
PC2 → PC6	18.18.1.11 → 18.18.2.11	DPC to PSC over the RIPv2 path through CCH, not the backup link	Reply
PC6 → PC8	18.18.2.11 → 18.18.3.130	PSC to the DAU stub site	Reply
PC4 → Web server	18.18.3.11 → 18.18.2.2	Reachability of the web server before testing it by name	Reply

10.2 Service tests

- **DNS and the web server.** From any PC open Desktop > Web Browser and request <http://www.transitline.local>. The page loading proves that the DNS server at CCH resolved the name and that the web server at PSC answered, so both the name service and the routing between the two centres are working.
- **DHCP.** On PC2, PC3, PC4, PC5, PC6, PC7 and the three DHCP printers, select **DHCP** and confirm the status reads *DHCP request successful*. The address should fall inside the right pool and the gateway and DNS server should be filled in automatically.
- **Email through the PSC relay.** Configure Desktop > Email on two PCs in different centres, then send a message from one to the other and press **Receive** on the second. Delivery proves that PSC is relaying mail across the routed network.

10.3 Commands to check the configuration

Table 11 — Verification commands and the expected result.

Router	Command	What to look for
Every router	show ip interface brief	All configured interfaces read <i>up / up</i> . A serial line showing <i>down / down</i> means the clock rate is missing on the DCE end.
Every router	show ip route	Confirms the routing table. Only RSP should report a gateway of last resort.
CCH, DPC, FDU, PSC	show ip protocols	Confirms RIPv2, <i>no auto-summary</i> , and that the passive interfaces are listed.
DPC	show ip route 18.18.2.0	Normally an R entry with distance 120. After the primary link is shut it becomes an S entry with distance 130 — the floating route.

Router	Command	What to look for
CCH	<code>show ip route rip</code>	Shows the routes RIPv2 contributes, including the DAU and RSP LANs that arrive through redistribution.
DPC, FDU	<code>show ip dhcp binding</code>	Lists the leases handed out by DPC_POOL and FDU_POOL and confirms they start at .11.
DPC, FDU	<code>show ip dhcp pool</code>	Confirms the pool size, the excluded range and how many addresses are leased.

10.4 Failure testing

The five scenarios in Table 8 complete the verification. Each one is produced by shutting a single interface, checking the routing table, running a ping, and then restoring the interface with `no shutdown` before moving on to the next.